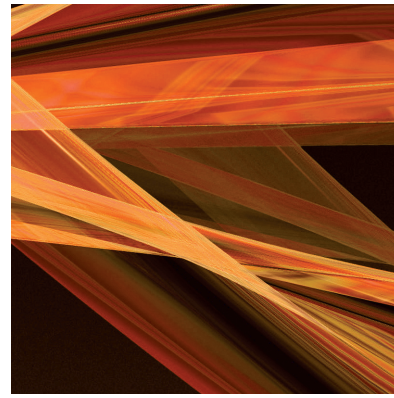


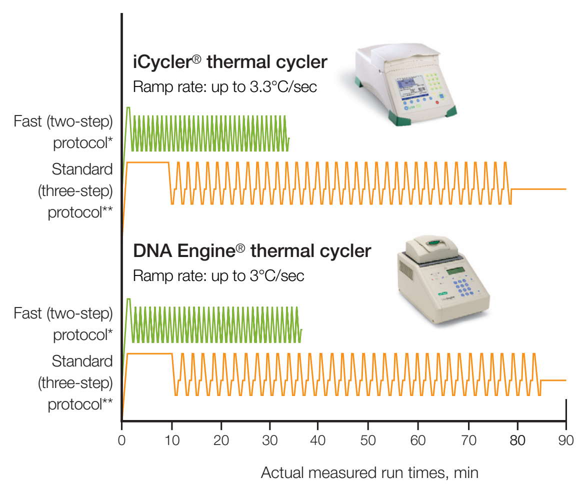
Thermal Cyclers and Reagents

Fast PCR With Bio-Rad Products

PCR run times can be dramatically reduced without giving up the flexibility, modularity, and gradient capability of Bio-Rad's thermal cyclers. With our cyclers, enzymes, and reaction vessels, you can:

- Run standard PCR reactions in only 35 minutes
- Obtain SYBR Green real-time PCR quantitation data in under 40 minutes
- Reduce run times for long (1–15 kb) targets 3- to 4-fold

For more information, visit us on the Web at **www.bio-rad.com/fastpcr/**



* Fast (two-step) protocol, 98°C for 30 sec, then (92°C for 1 sec, 70°C for 20 sec) x 35 cycles.

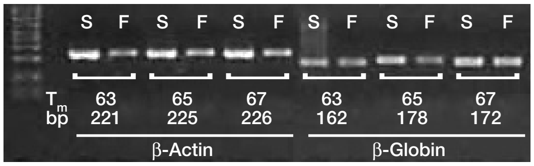
** Standard (three-step) protocol, 95°C for 10 min, then (95°C for 15 sec, 60°C for 30 sec, 72°C for 30 sec) x 35 cycles, 72°C for 10 min.

Easy Protocol Conversion Is the Key to Dramatically Reduced Run Times

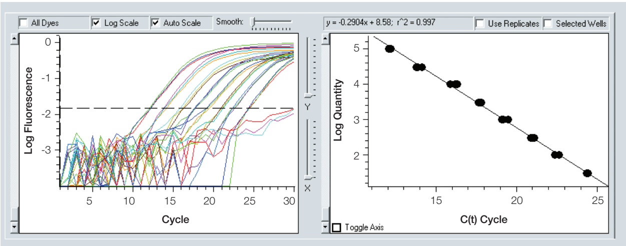
With standard Bio-Rad thermal cyclers, enzymes, and reaction vessels, a 1.5 hour PCR protocol can be reduced to 35 minutes:

- Shorten initial hot-start step with iTaq™ polymerase
- Combine annealing and extension steps at 70ºC
- Reduce hold times for annealing/extension
- Ensure primer T_m values are close to 65ºC

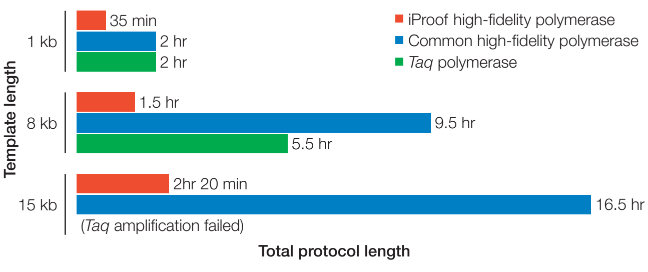
For details, download our fast PCR tutorial at **www.bio-rad.com/fastpcr/**



Fast two-step protocol results are comparable to those generated using standard protocols. β-Actin and β-globin targets were amplified from human genomic DNA using iTaq polymerase. Standard (S) protocol, 1.5 hr, fast (F) protocol, 35 min. For details see **www.bio-rad.com/fastpcr/**



SYBR Green qPCR in less than 40 min. Lambda phage target (150 bp) was amplified using iTaq™ SYBR® Green supermix. For details see **www.bio-rad.com/fastpcr/**



For long (1–15 kb) targets, use of iProof™ polymerase reduces run times 3- to 4-fold. Targets of 1, 8, or 15 kb were amplified using three different polymerases. A two-step PCR protocol was used with iProof polymerase; three-step protocols using the shortest recommended extension times were used with other polymerases. Because iProof polymerase requires an annealing temperature 5°C above typical annealing temperatures, two-step protocols often can be run without redesigning primers.

Ordering Information

Catalog #	Product Description
PTC-1148C	**MJ Mini™ Gradient Thermal Cycler With Adjustable Heated Lid,** holds 10 x 0.5 ml tubes, 48 x 0.2 ml tubes, or 48-well microplate; includes 120 x 0.2 ml 8-tube strips, 120 optical flat 8-cap strips, 1,000 x 0.2 ml individual tubes with attached flat caps
PTC-0200G	**DNA Engine Thermal Cycler Chassis**, does not include the Alpha™ unit (requires 1)
PTC-0220G	**DNA Engine Dyad® Dual-Bay Thermal Cycler Chassis,** does not include the Alpha units (requires 2)
PTC-0221G	**Dyad Disciple™ Dual-Bay Thermal Cycler Chassis,** does not include the Alpha units (requires 2)
PTC-0240G	**DNA Engine Tetrad® 2 Thermal Cycler Chassis,** does not include the Alpha units (requires at least 2 and fits up to 4)
170-9703	**MyCycler™ Thermal Cycler System,** includes MyCycler thermal cycler, PCR tubes, iTaq DNA polymerase, power cord, quick reference guide, instructions
170-8720	**iCycler Thermal Cycler With 96-Well Reaction Module**
170-8722	**iCycler Thermal Cycler With 2 x 48 Dual Reaction Module**
CFB-3120G	**MiniOpticon™ Real-Time PCR Detector,** includes optical housing, thermal cycler, and analysis software
CFB-3240	**Chromo4™ Four-Color Real-Time PCR Detection System,** includes optical housing, photonics shuttle, DNA Engine thermal cycler, 96-well sample block, analysis software
170-9770	**MyiQ™ Single-Color Real-Time PCR Detection System,** includes iCycler base, optics module, software CD-ROM, 96-well optical reaction module, optical-quality 96-well PCR plates, Microseal 'B' seals, communication cables, power cords, instructions
170-9780	**iQ™5 Multicolor Real-Time PCR Detection System,** includes iCycler base, optics module, software CD-ROM, 5 installed filter sets, 96-well reaction module, calibration solutions, optical-quality 96-well PCR plates, Microseal 'B' seals, communication cables, power cord, quick reference cards, instructions
170-8870	**iTaq DNA Polymerase,** 5 U/µl, includes 250 U polymerase, 1.25 ml 10x PCR buffer, 1.25 ml 50 mM $MgCl_2$ solution
172-5301	**iProof High-Fidelity DNA Polymerase,** 2 U/µl, includes 100 U enzyme, 5x reaction buffers, $MgCl_2$ solution, DMSO
172-5302	**iProof High-Fidelity DNA Polymerase,** 2 U/µl, includes 500 U enzyme, 5x reaction buffers, $MgCl_2$ solution, DMSO
172-5310	**iProof HF Master Mix,** 100 x 50 µl reactions, includes 2x master mix, DMSO (for highest fidelity with most templates)
172-5311	**iProof HF Master Mix,** 500 x 50 µl reactions, includes 2x master mix, DMSO (for highest fidelity with most templates)
170-8880	**iQ™ SYBR® Green Supermix,** 100 x 50 µl reactions
170-8882	**iQ SYBR Green Supermix,** 500 x 50 µl reactions

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